

**AGENTIC ENGINEERING ASSESSMENT · SOFTWARE
EMPHASIS**

External signals × workforce retention

Asteria Consumer Products (fictional). How three retention targets are performing, and whether the economy adds useful context.

One command, runs offline

5 public signals, 2 providers

128 automated tests

Every number re-derived: 107 / 107

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The question, narrowed

Decision supported: **which retention target needs leadership attention, and does the economy explain it?**

TARGET	GOAL	HOW I DEFINED IT	OPEN QUESTION → ASSUMPTION
New hires kept 6 months	≥ 86%	Hires grouped by month from Jan 2021; "lost" = left before the 6-month anniversary	Calendar months; unfinished groups are <i>pending</i> , never "retained"
Senior hires kept 12 months	≥ 90%	Same rule, 12 months, Senior Leader only	Who is senior? → Senior Leader (+ "Sr Mgmt" alias); Managers as a sensitivity check
Regretted turnover, trailing 12 months	≤ 7.5%	Regretted exits ÷ average of the 12 month-end headcounts	Denominator convention → mean of month-end headcounts

Clarification round

10 candidate questions drafted, cut to the **3 that can flip a result**; the other 7 recorded as assumptions. Every assumption is a switch in `config/analysis.yaml`.

Deliberately out of scope

Forecasting, causal claims, individual-level drivers (pay, manager), per-country publication lags. **Depth over surface area.**

The data is deliberately imperfect: nothing is dropped silently

23 quality rules, each with a severity that decides the row's fate. Counts reconciled with an independent profiling script.

ISSUE FOUND	ROWS	WHAT HAPPENED
Exact duplicates	7	Removed
Country "EL" / "ROM"	8	Mapped to GR / RO (EL is Eurostat's own code)
Blank country	9	Company-wide figures only
"Sr Mgmt" level	10	Mapped to Senior Leader (assumption)
No hire date / left before hired	5 + 5	Excluded from all measures
Exit with no exit type	13	Counted as an exit, never as regretted
"End of Contract" on a Permanent	90	Kept, flagged
Voluntary exit, regretted unknown	2	Main: not regretted; upper bound: regretted

2,407 → 2,400 → 2,390

rows read → people → measurable. 2,381 have a known country.

The trap: recent hires

Someone hired in October 2025 hasn't had 6 months yet. Counting them as "stayed" would inflate retention, so their group is **pending**. That's why every hire-based headline covers **2021–2024**.

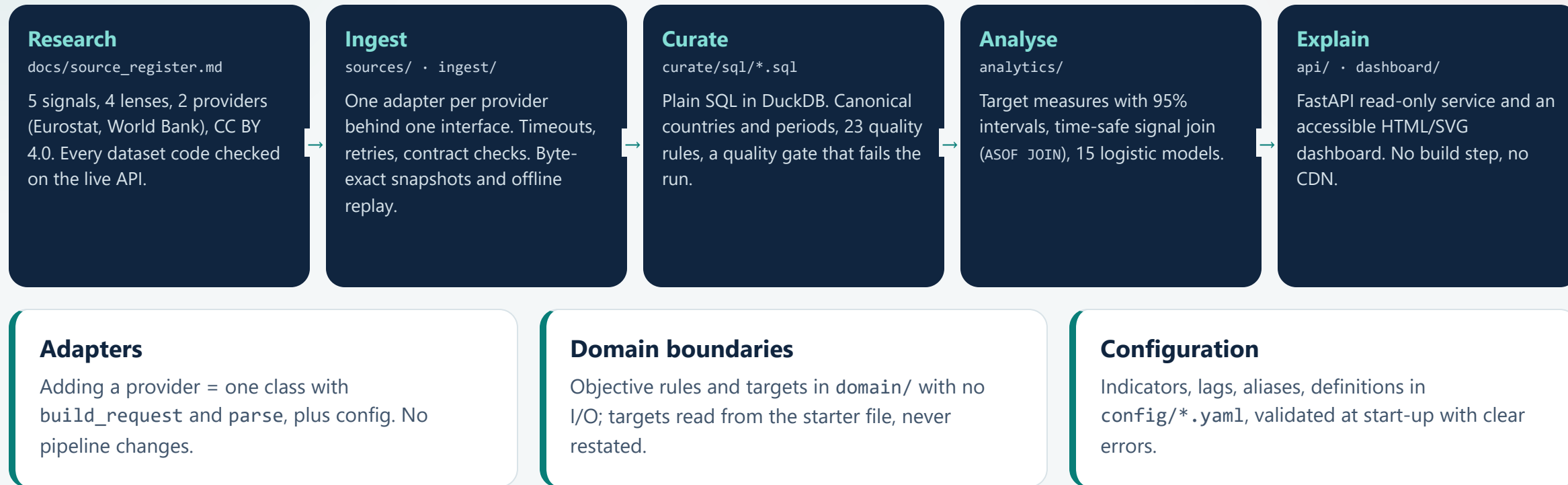
Also found

Nobody was hired before 2020, so headcount grows from 0 to ~1,600, which matters for reading turnover trends.

IN PLAIN WORDS We never quietly delete a messy row: each problem has a rule, and every rule's count is published.

One vertical slice, one command: `uv run asteria run`

Each stage is a separate module with its own tests; the core runs offline from committed snapshots in about 20 seconds.



Lineage: every value can be traced back to its source bytes

LAYER	WHERE	WHAT IT HOLDS	PROOF IT'S FAITHFUL
Raw	data/raw-or-fixtures/	Starter CSVs + API payloads, exactly as received	SHA-256 per file, checked by tests; <code>.gitattributes</code> stops line-ending edits
Source-shaped	DuckDB stg_*	Provider codes and period strings, untyped	Parsed only after a contract check (schema drift is an error)
Canonical	employees, external_observations	ISO countries, real period dates, quality flags	Lineage columns: source, dataset, period, loaded_at, snapshot sha
Analytical product	mart_*	Rates + intervals per country × unit × period; signals as known at the time; model results	Recomputed by separate code: 107 / 107 match

Deterministic

Reruns are **byte-identical**, including from a clean clone.

Idempotent

A second live ingest reports every source **unchanged**; nothing is rewritten.

Schema drift caught

Eurostat renamed `nace_r2` to `nace_r2_1`: the adapter treats a missing dimension as a contract error.

IN PLAIN WORDS For any number on the dashboard we can point to the exact downloaded file, and the exact rows, it came from.

No future information: a value is used only after it was published

SIGNAL	LAG	REAL RELEASES (SPOT CHECK)
Unemployment	62 d	30–33 days after month end
Job vacancies	90 d	75–79 days after quarter end
Inflation	25 d	17–24 days (January latest)
Economic sentiment	10 d	Before month end, except December: 5–8 Jan
GDP growth	200 d	Enters World Bank data on 1 July

Leak found and fixed: sentiment originally had a 0-day lag, so December values were used from 1 January, 5–8 days before publication (4,861 person-months). Fixed to 10 days; the conclusion didn't change.

What Greece "knew" on 1 June 2023

SIGNAL	VALUE	DESCRIBES	AGE
Sentiment	108.0	Apr 2023	32 d
Unemployment	11.4%	Mar 2023	62 d
Vacancies	0.9%	Q4 2022	152 d
GDP growth	8.7%	year 2021	517 d

Frequency integrity: an annual figure stays **labelled as its own year and age**, never shown as a monthly measurement.

Why it matters: the naive "same month" join produced the **strongest-looking result (p = 0.010)**. Leakage makes weak patterns look real.

IN PLAIN WORDS The analysis may only use economic news that had actually been published by that date, like reading yesterday's paper, never tomorrow's.

Reliability and engineering

Failure handling

- Transient errors (timeouts, 429, 5xx) retried with backoff; 4xx fail fast with the provider's message
- One failing source never stops or corrupts the others; its last good snapshot is kept
- A payload that breaks the contract never replaces good evidence
- API error contract: 404 / 422 (lists allowed values) / 503 "run asteria run" / 500 with request id, no internals

Observability

- Readable logs in a terminal, JSON when piped or scheduled
- Run manifest per ingest: attempts, checksums, coverage, warnings
- Quality report with every rule and its row count
- Dashboard "Trust" view: freshness, gaps, provider flags

TEST LAYER	WHAT IT PROVES
Unit + contract	Parsers, JSON-stat decoding, retries, raw store, config validation
SQL / metrics	Hand-computed cases: anniversary boundary, pending, headcount, Wilson interval, no-leak and period rules
Statistics	Known-answer tests: finds a planted effect; not fooled by country confounding
API + UI	Every endpoint and error; Playwright browser tests incl. empty/error states; axe accessibility audit in both themes

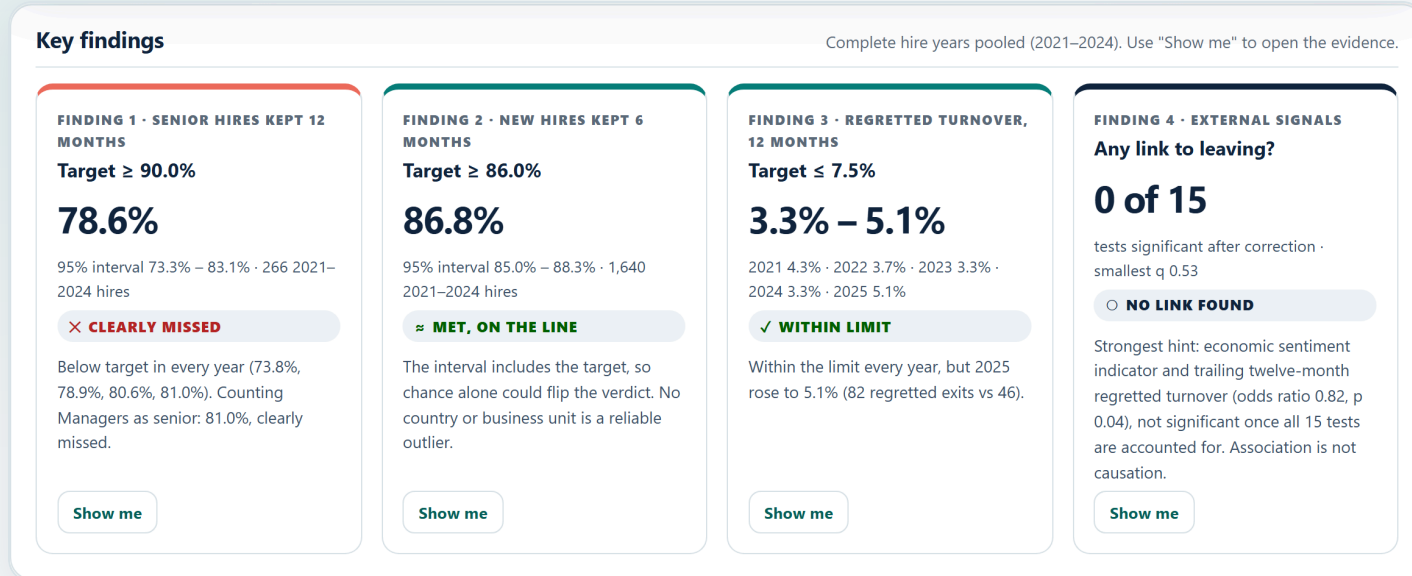
128 tests on Python 3.13 and 3.11 · strict mypy · ruff · tests validated by injecting bugs on purpose (each was caught)

Production mapping: ADF · Databricks · Power BI

CONCERN	LOCAL (BUILT)	PRODUCTION (PROPOSAL)
Orchestration	asteria run	ADF pipeline: ingest → Databricks jobs (curate, analyse) → Power BI refresh
Scheduling	Manual	Monthly trigger after the latest release + event trigger on a new HR extract
Storage	Files + DuckDB	ADLS Gen2, bronze / silver / gold Delta tables in Unity Catalog
Secrets	None needed	Azure Key Vault via managed identities; Key Vault-backed Databricks secret scopes
Observability	JSON logs, manifest, quality report	Log Analytics; manifest and quality as Delta tables; alerts on failure, gate, staleness
Access control	Local	Unity Catalog grants (person-level HR restricted); Power BI row-level security by country
Promotion	Git	Dev → Test → Prod; Databricks Asset Bundles + ADF ARM; CI runs ruff, mypy, pytest per PR

Carries over unchanged: adapters, domain rules, SQL, config, tests. Changes for real HR data: pseudonymised IDs, incremental MERGE instead of full rebuild, retention policy. Details: docs/architecture.md.

The dashboard: findings first, then prove them



Explore

Objective, country, business unit, period, definition, signal.
Kept in the URL.

Understand

Trend with 95% band and target line, next to the signal as known each month.

Challenge

Country-quarter scatter sized by sample; naive vs within-country tests.

Trust

Freshness, coverage, exclusions, sources, method.

Accessible by design: keyboard-readable charts, a table for every chart, status as icon + text, validated colour-blind palette, graceful empty / pending / "too few people" / API-down states.

Reading the numbers, in plain words

Every figure on the next slides is one of these. Example used throughout: senior hires in 2024.

Rate	Of the 63 senior people hired in 2024, 51 were still here 12 months later: $51 \div 63 = \mathbf{81.0\%}$.
Pooled	Add the complete years together (people, not percentages): 209 stayed of 266 hired in 2021–2024 = 78.6% .
95% interval	How sure we are. With only 63 people the true rate could plausibly be anywhere from 69.6% to 88.8% ; more people give a narrower range.
"Clearly" vs "on the line"	Clearly : the whole range is on one side of the target. On the line : the range includes the target, so chance alone could flip the verdict.
Pending	Too early to judge: hired less than 6 or 12 months before the data ends. Never counted as "stayed".
Too few people	Fewer than 30 people in a group: shown, but not judged. One person leaving would move the rate by more than 3 points.
Turnover	Regretted exits in a year $\div$ the average of that year's 12 month-end headcounts: $82 \div 1,609.7 = \mathbf{5.1\%}$ in 2025.

Findings 1–3: how the targets are doing

Finding 1 · senior hires kept 12 months · target $\geq 90\%$

78.6%

95% interval 73.3–83.1% · 209 of 266 hires, 2021–2024

✗ **CLEARLY MISSED**

- Below target **every year**: 73.9, 78.9, 80.6, 81.0%
- Counting Managers as senior: 81.0%, still clearly missed
- Largest and most robust gap

Finding 2 · new hires kept 6 months · target $\geq 86\%$

86.8%

95% interval 85.0–88.3% · 1,423 of 1,640 hires

≈ **MET, ON THE LINE**

- Every year's interval includes 86%
- Countries 84.7–89.2%, units 85.1–88.8%: no reliable outlier
- Read "met" as "on the line"

Finding 3 · regretted turnover · limit $\leq 7.5\%$

3.3–5.1%

4.3 · 3.7 · 3.3 · 3.3 · **5.1%** (2021–2025)

✓ **WITHIN LIMIT, RISING**

- 2025: 82 regretted exits vs 46 in 2024
- Romania 6.7%, Ireland 6.2%, Bulgaria 6.1%: closest to the limit
- Caveat: young, growing workforce may explain part

How to read the verdicts: "**clearly**" means the whole 95% interval is on one side of the target (for senior hires even the top of the interval, 83.1%, is below 90%). "**On the line**" means the interval includes the target, so chance alone could flip it.

The economic question, in plain words

3 objectives = 3 different kinds of leaving

New hires leaving early, senior hires leaving early, and regretted voluntary exits. We test each separately, because the economy can push them in different directions.

5 signals × 3 objectives = 15 questions

Unemployment, job vacancies, inflation, economic sentiment and GDP growth, each tested against each objective.

ONE OF THE 15 QUESTIONS

"When unemployment is higher than usual in a country, are employees there more likely to have a regretted exit?"

Why we need leavers and stayers

Looking only at the people who left tells us what the economy looked like *when they left*, but not whether it made leaving **more likely**. For that we compare everyone at risk at the same moment: 65,184 person-months for turnover.

Three terms you'll hear

- **Odds ratio (OR):** 1.00 = no link; 0.82 = 18% lower odds of leaving per typical rise in the signal
- **p:** "could this one result be luck?" (small = unlikely luck)
- **q:** the same, after allowing for asking 15 questions. We need q below 0.05

Testing *any* exit is possible as an extra exploratory check, but it answers a broader question than the three objectives, so it isn't part of the findings.

THE KEY IDEA We ask whether changes in external conditions go together with changes in the likelihood of a specific type of leaving, comparing each country only with itself.

Finding 4: no economic signal lines up with leaving

0 of 15

tests significant after correcting for running 15 tests · smallest q = **0.53**
(needs < 0.05)

Method

- One row per person (or person-month), not averages
- Each country compared only with itself, plus a time trend
- Errors clustered by country-month; Benjamini-Hochberg correction

Why trust a "no"

- Same code finds a planted effect in synthetic data
- 8 alternative setups tested (timing, lags, controls, errors): none changed the answer
- Pooling countries flips unemployment's direction (0.97 → 1.11): that's the confounding the method removes

Regretted exits per person-month vs economic sentiment indicator

One dot per country and quarter (calendar quarter); signal as known at the time.



Small dots at 0% or 100% are quarters with only a handful of people; the model weights every person equally, so they do not drive the result.

► Show data table

Is there an association?

Outcome: regretted exit in the month (per person-month at risk).
Odds ratio per one typical within-country swing of the signal; 1.00 means no effect. "Within country" compares each country only with itself and removes a common time trend.

SIGNAL	NAIVE OR (95% CI)	WITHIN COUNTRY OR (95% CI)	Q	VERDICT
Economic sentiment indicator	0.87 (0.77–1.00)	0.82 (0.68–0.99)	0.53	Nominal only; not significant after correction
GDP growth	0.90 (0.79–1.02)	0.89 (0.77–1.02)	0.65	No evidence
HICP all-items	0.91 (0.80–1.04)	0.89 (0.75–1.05)	0.77	No evidence
Job vacancy rate	0.96 (0.88–1.05)	0.93 (0.78–1.12)	0.80	No evidence
Unemployment rate	0.97 (0.91–1.04)	1.11 (0.84–1.45)	0.80	No evidence

Only hint: higher business confidence, fewer regretted exits (odds ratio 0.82, p = 0.04).
With 15 tests, a p this small is expected by chance, so q = 0.53. **No evidence of an effect is not evidence of no effect:** 57 senior leavers is a small sample.

How the conclusion was reached

ASTERIA RETENTION SIGNALS · HOW THE CONCLUSION WAS REACHED

From raw data to four findings

Left: the employee path (findings 1–3 use only this). Right: the external path, joined to employee outcomes to test finding 4.

1 INPUTS

Workforce file (synthetic)

2,407 rows: hires and exits across 6 countries, 2020–2025.
3 targets from the starter pack.

External data (public)

5 signals, 2 providers: unemployment, job vacancies, inflation, economic sentiment (Eurostat); GDP growth (World Bank).
2,456 values, monthly, quarterly and annual.

2 PREPARE

Clean and flag (23 quality rules)

- 7 duplicates removed → 2,400 people
- 10 invalid records excluded → 2,390 measurable
- 9 with no country: company-wide figures only

Time-safe join ("what was known then")

- A value is used only after its publication date
- Lags: 62 / 90 / 25 / 10 / 200 days, checked against real release dates
- Annual values keep their own year and age

3 MEASURE AND TEST

Measure the three targets

- Group hires by month; lost = left before the 6- or 12-month anniversary
- Unfinished groups are pending, never counted as retained
- Turnover = regretted exits ÷ average of 12 month-end headcounts
- 95% interval on every rate; fewer than 30 people = not judged

Test: does the economy line up with leaving?

- 15 models (3 targets × 5 signals), one row per person
- Each country compared only with itself, plus a time trend
- Correction for running 15 tests (Benjamini-Hochberg q)

↗ employee outcomes feed the test ↘

4 FINDINGS (POOLED 2021–2024 HIRES; TURNOVER PER CALENDAR YEAR)

FINDING 1 · SENIOR HIRES KEPT 12 MONTHS
Target ≥ 90%

78.6%

95% interval 73.3–83.1% · 266 hires

✗ CLEARLY MISSED

Every year below target (73.9 → 81.0%). Still missed if Managers count as senior (81.0%).

FINDING 2 · NEW HIRES KEPT 6 MONTHS
Target ≥ 86%

86.8%

95% interval 85.0–88.3% · 1,640 hires

≈ MET, ON THE LINE

Every year's interval includes 86%. No country or business unit is a reliable outlier.

FINDING 3 · REGRETTED TURNOVER, 12 MONTHS
Limit ≤ 7.5%

3.3–5.1%

4.3 · 3.7 · 3.3 · 3.3 · 5.1% (2021–2025)

✓ WITHIN LIMIT, RISING

2025 up (82 exits vs 46). RO, IE, BG closest to the limit. Growing, young workforce may explain part.

FINDING 4 · EXTERNAL SIGNALS
Any link to leaving?

0 of 15

significant after correction · smallest q 0.53

○ NO ASSOCIATION

Only hint: higher sentiment, fewer regretted exits (p 0.04), not significant after correction. Association ≠ causation.

107 / 107

every number recomputed by separate code

124 tests

metrics, time rules, API, dashboard, accessibility

Lags spot-checked

against official release dates; one leak found and fixed

Byte-identical

rebuild from a clean clone

Assumptions (pending clarification): senior = Senior Leader (Sr Mgmt mapped to it); 2021+ monthly hire cohorts; turnover denominator = mean of 12 month-end headcounts. Workforce data is synthetic.

Tradeoffs and limitations

Limits of the evidence

- Synthetic, hiring-only workforce (nobody before 2020): turnover trends mix real change with a maturing workforce
- Six countries, about five years: only moderate-to-large effects detectable
- The turnover model doesn't control for time since hire, which strongly drives leaving (2.4% a month in the first 6 months vs 0.5% after 2 years)
- Definitions rest on assumptions until the clarification answers arrive (each is a config switch)
- Only inflation has first-release values; others are today's revised figures (inflation checked: never revised)
- Italy's vacancy rate uses a different definition (Eurostat flag)

Deliberate choices

- DuckDB + FastAPI + plain HTML: free, offline, easy to review; maps cleanly to Databricks + Power BI
- Conservative lags: can only make data slightly stale, never leak
- One lag per series (Ireland publishes a month earlier; not exploited)
- Logistic regression over machine learning: the question is "is there a link?", not "predict who leaves"
- Full rebuild instead of incremental loads, which is right at 2,400 rows

How I used the AI agent, and stayed accountable

Decisions that stayed mine

- Software emphasis; repository shape (changed the agent's layout to the brief's)
- Cut clarification questions 10 → 3; analyse blind, not from the data generator
- Indicator set (from my own search, fact-checked by the agent)
- Methodology: when I found I hadn't been consulted, every option was put back to me and re-run
- Lag fixes, dashboard changes, pooling rule

Agent errors caught by verification

- Defect list from reading code: 3 overstatements, 1 missed; fixed by profiling
- Two licence URLs recalled from memory were wrong
- 0-day sentiment lag leaked December data; found by the release-date spot check
- Interval formula crashed on tiny slices; found by its own test
- Accessibility contrast failures; found by the axe audit

Verification, not trust: tests, an independent recomputation of every number (107 / 107), live checks against provider APIs and release dates, screenshots reviewed. All logged in `AI_USAGE.md`, including my own corrections.

Next steps

#	NEXT STEP	WHY
1	Apply the clarification answers	Change config, rerun: senior scope and the headcount rule
2	Act on senior retention	The clear gap; look at internal drivers (onboarding, role fit)
3	Release-calendar ingest	Exact publication dates and per-country lags instead of fixed lags
4	Point-in-time vintages	First-release values for every signal, not only inflation
5	Robustness checks	Add time since hire to the turnover model; linear-probability and survival models alongside logistic
6	Productionise	ADF + Databricks + Power BI per the mapping; CI/CD, alerts, RLS

Effort

Approximately **[fill in] hours** across framing, sources, pipeline, analysis, dashboard, and documentation.

Try it

```
git clone .../asteria-retention-signals
uv sync · uv run asteria run · uv run asteria serve
```

Read more

[docs/findings.md](#) · [docs/analysis_explained.md](#) · [docs/source_register.md](#) · [docs/architecture.md](#) · [AI_USAGE.md](#)